

BEYOND THE OPTIMUM

*Chemical Engineering Decisions
with Multiple Objectives and Uncertainty*

*From a single “best” answer
to better engineered decisions*

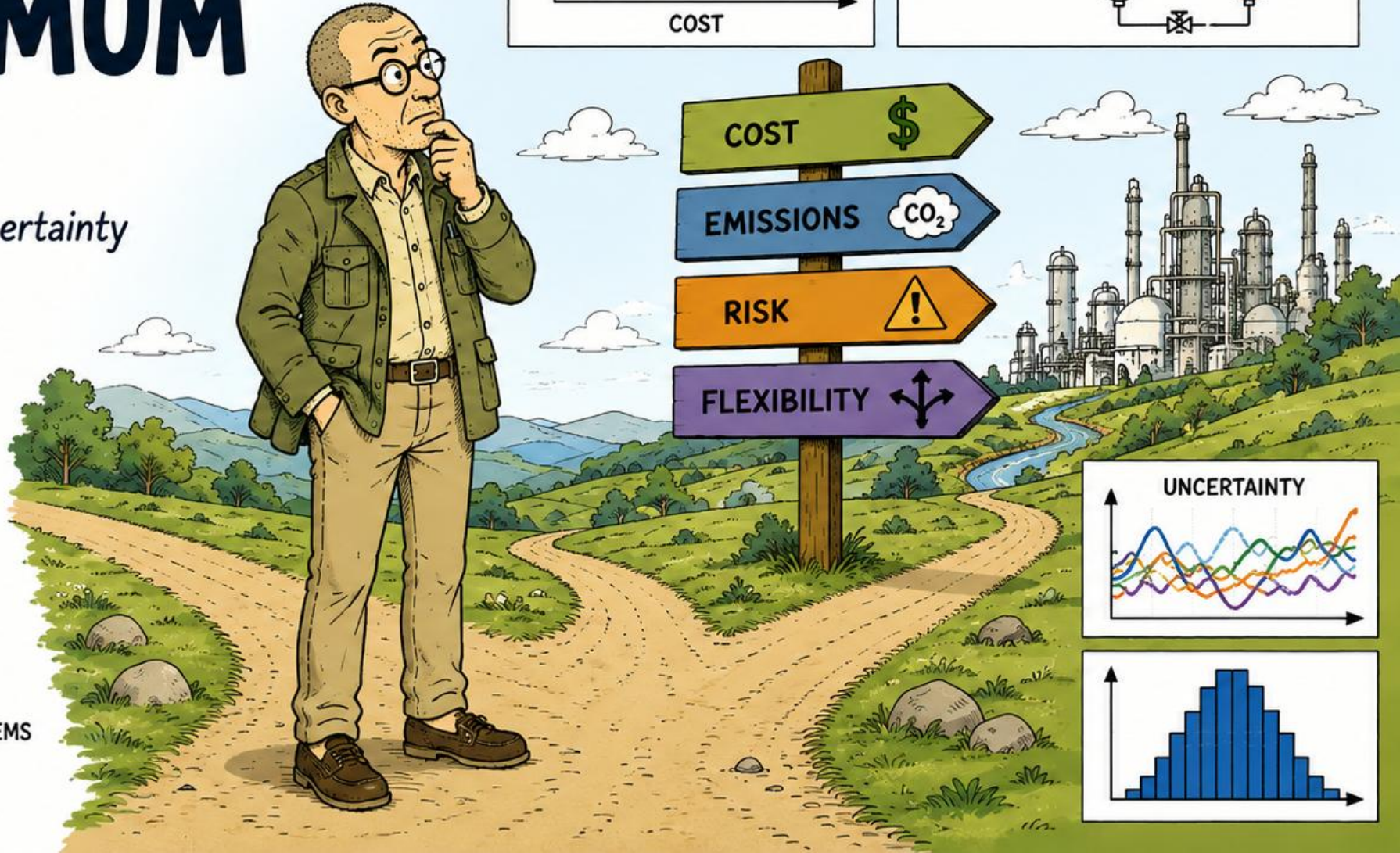
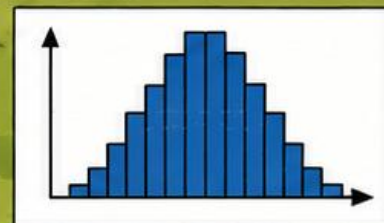
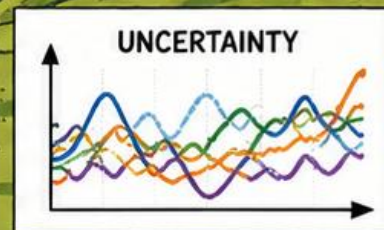
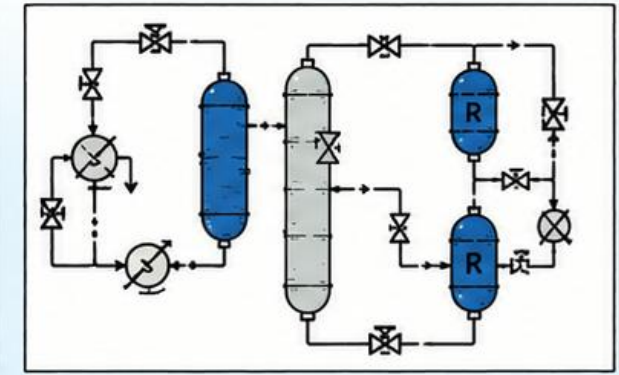
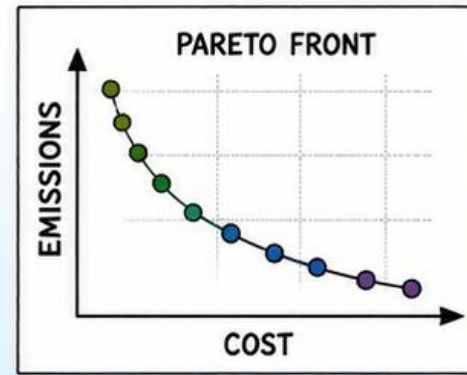
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UNIVERSITY
OF TWENTE.

PSE

PROCESS SYSTEMS
ENGINEERING



WHAT SHOULD WE BUILD?

Suppose we want to produce *methanol*.

We have carbon dioxide (CO_2).

We need hydrogen (H_2).

We combine them in a synthesis process, separate the products, recycle what we can, and produce methanol (CH_3OH).



There are many ways to do this.

Which process should we build?

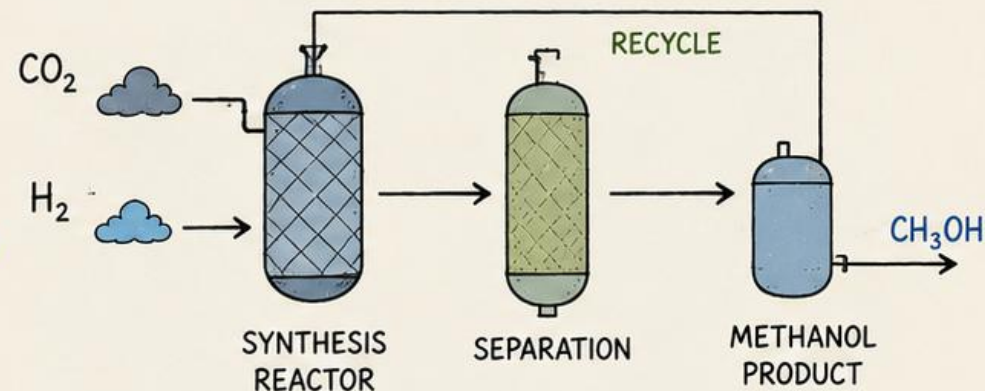


Our goal today:

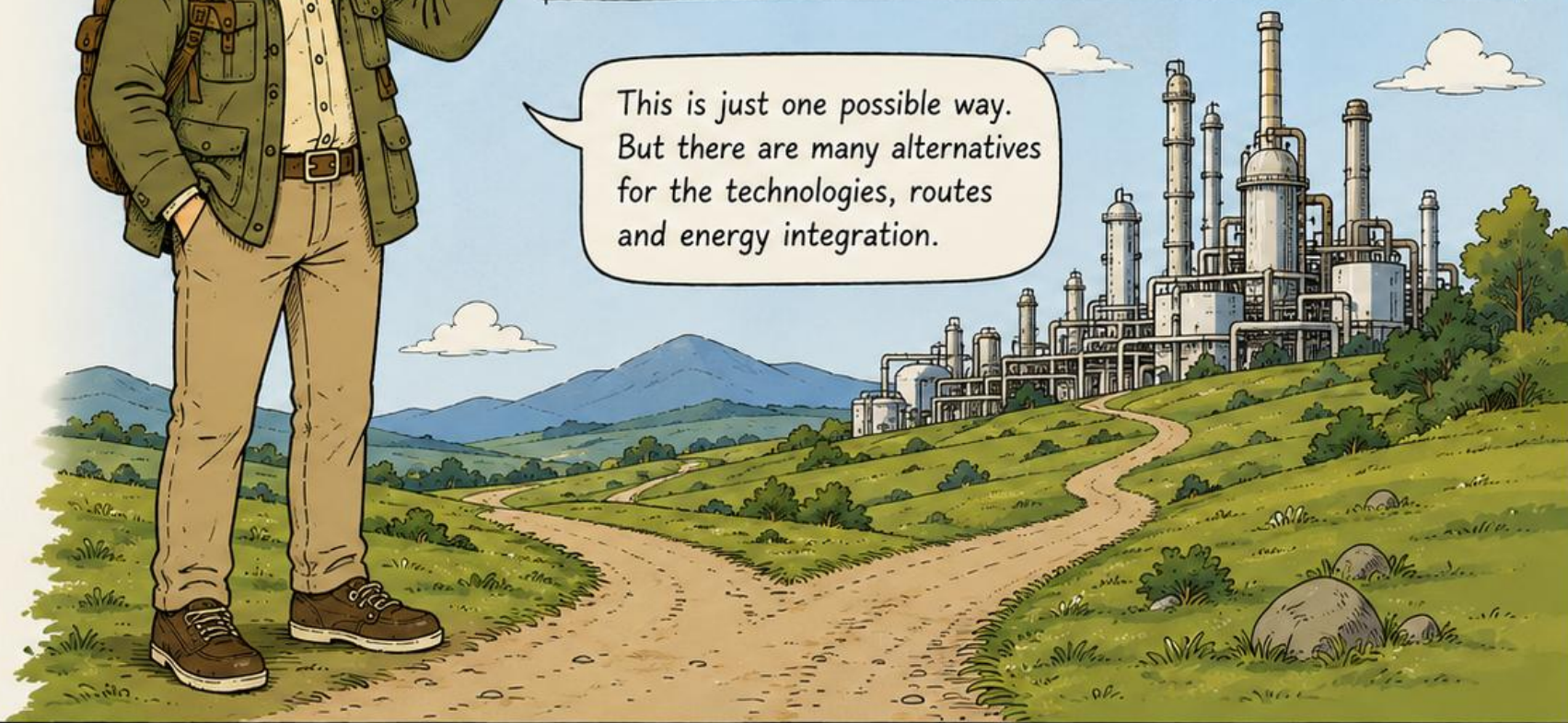
Better engineered decisions

with multiple objectives and uncertainty.

A SIMPLE METHANOL PROCESS (SCHEMATIC)



This is just one possible way.
But there are many alternatives
for the technologies, routes
and energy integration.



EASY. OPTIMIZE IT.

We build a model of the process.
We describe the balances, the thermodynamics, the equipment, and the operating variables.

Then, being engineers, we formulate an optimization problem.

Let's optimize this process.
Simple!

OPTIMIZATION PROBLEM

Minimize C_{ann} (annualized cost)

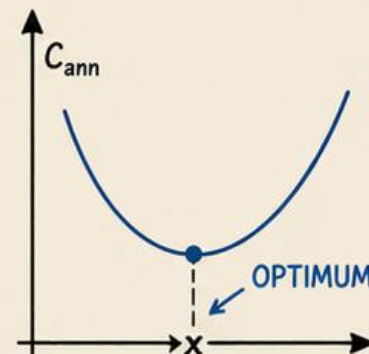
Subject to

Mass and energy balances

Process constraints

Bounds on variables

$$x_L \leq x \leq x_U$$



DECISION VARIABLES (EXAMPLES)

- Reactor temperature
- Pressure
- Recycle ratio
- Equipment sizes
- Heat integration level

x

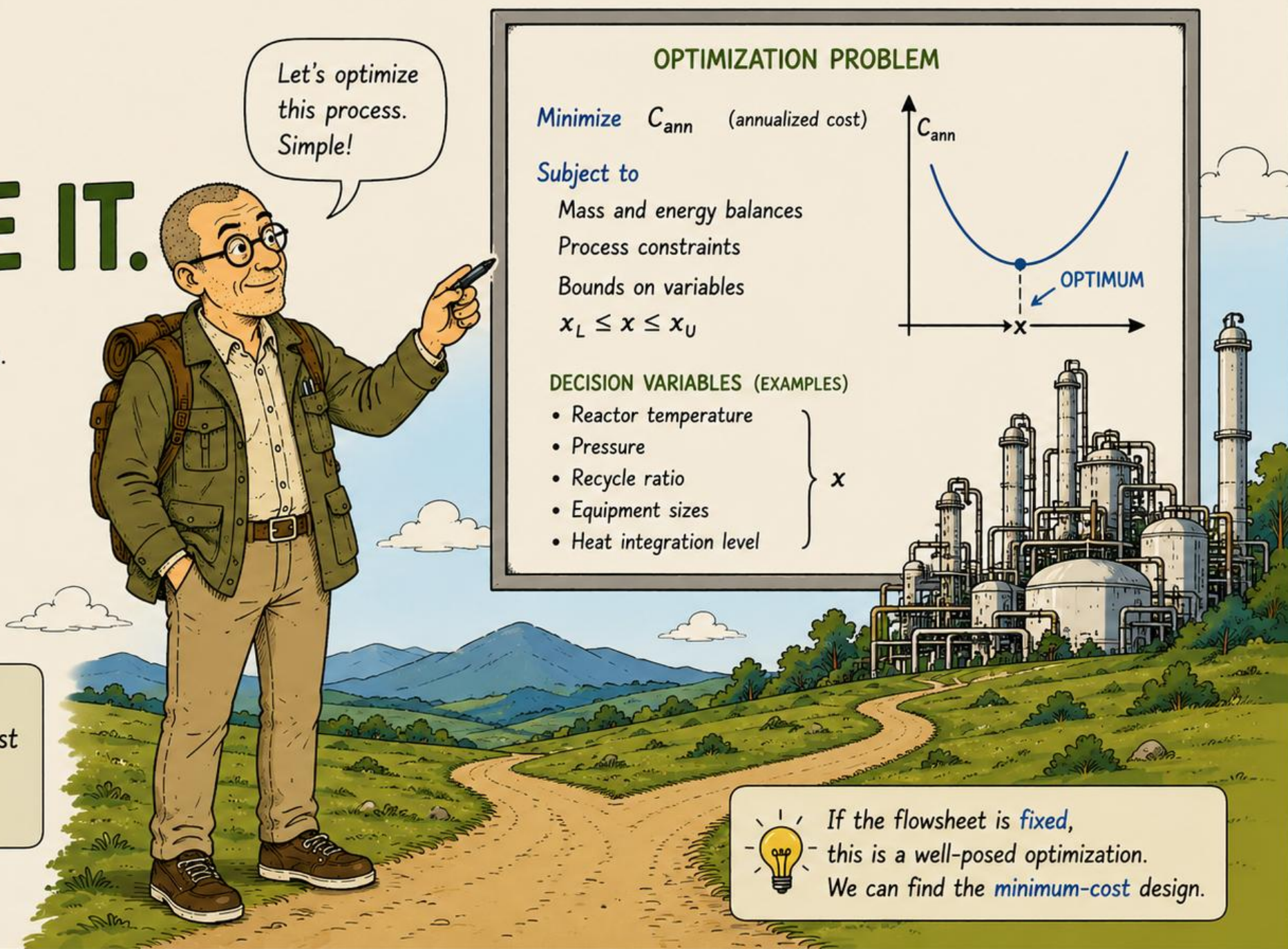


Objective:

Minimize annualized cost while satisfying all process constraints.



If the flowsheet is *fixed*, this is a well-posed optimization. We can find the *minimum-cost* design.



BUT WHICH PROCESS?

There are many ways to produce *methanol*.

Different choices for:

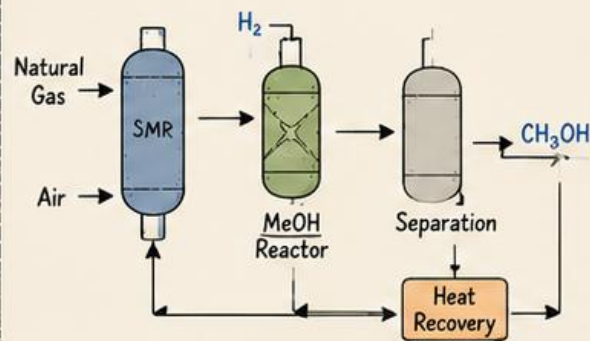
- Reactor
- H_2 source
- Energy integration
- Separation
- Heat recovery

Which flowsheet should we optimize?

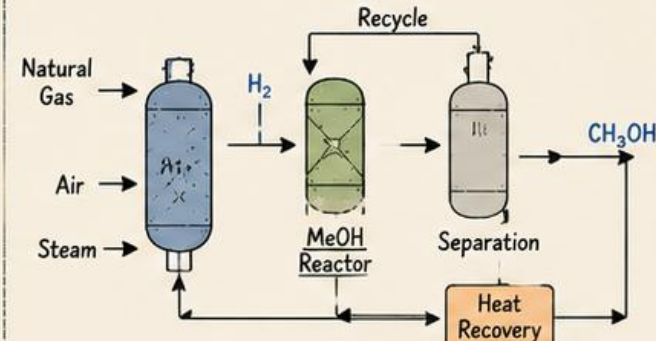
Many alternatives... each with different choices and consequences.

MULTIPLE ALTERNATIVE FLOWSHEETS FOR METHANOL PRODUCTION

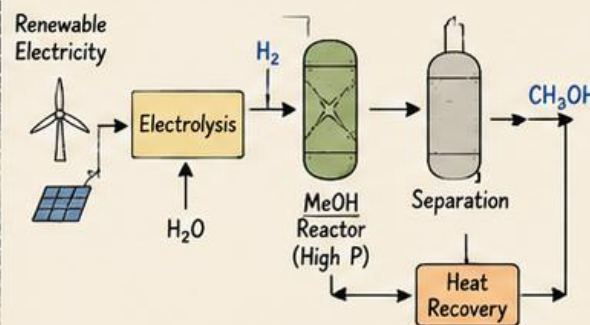
A. SMR + CONVENTIONAL REACTOR



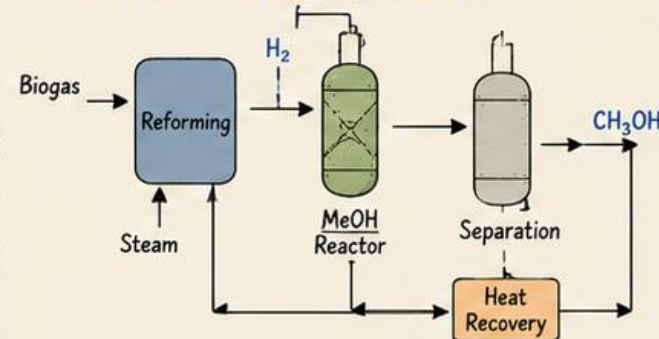
B. ATR + REACTOR WITH RECYCLE



C. ELECTROLYSIS H_2 + HIGH PRESSURE REACTOR



D. BIOGAS + REFORMING + REACTOR



Different choices for each building block

REACTOR OPTIONS



- Fixed bed
- Slurry bubble column
- Isothermal
- Adiabatic

H_2 OPTIONS



- SMR
- ATR
- Electrolysis
- Biomass / Biogas

ENERGY OPTIONS



- Heat integration
- Heat pumps
- Power generation
- Pinch design

SEPARATION OPTIONS



- Distillation
- Membrane
- PSA
- Hybrid

HEAT RECOVERY OPTIONS



- Recuperation
- Heat exchange
- Steam generation
- Thermal storage



Different choices lead to very different performance in cost, emissions, safety and energy use.

DON'T CHOOSE YET. PRESERVE THE ALTERNATIVES.

Many different flowsheets can achieve *methanol* production.

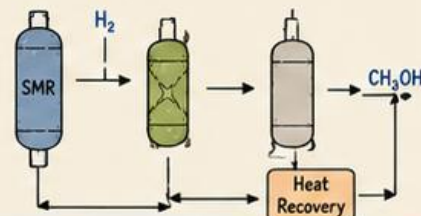
Instead of choosing now, we merge all reasonable alternatives into one *superstructure*.

The *optimization* will decide which path is best.

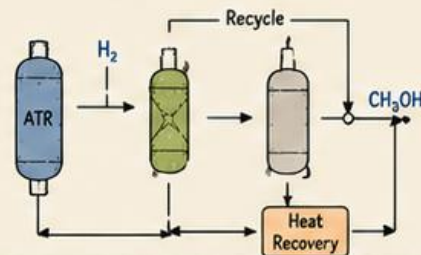
Don't commit too early.
Keep your options open.

MANY ALTERNATIVE FLOWSHEETS

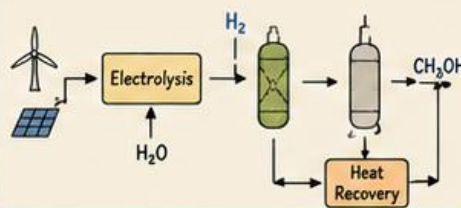
A. SMR + CONVENTIONAL REACTOR



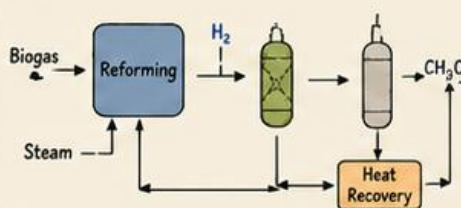
B. ATR + REACTOR WITH RECYCLE



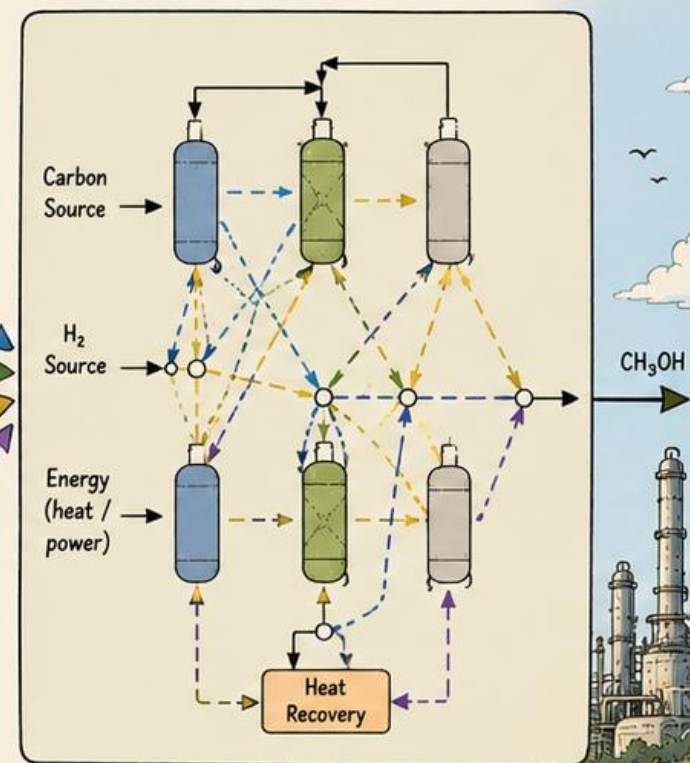
C. ELECTROLYSIS + HIGH PRESSURE REACTOR



D. BIOGAS + REFORMING + REACTOR



ONE SUPERSTRUCTURE



Optimization selects the best path through the superstructure under all objectives and constraints.

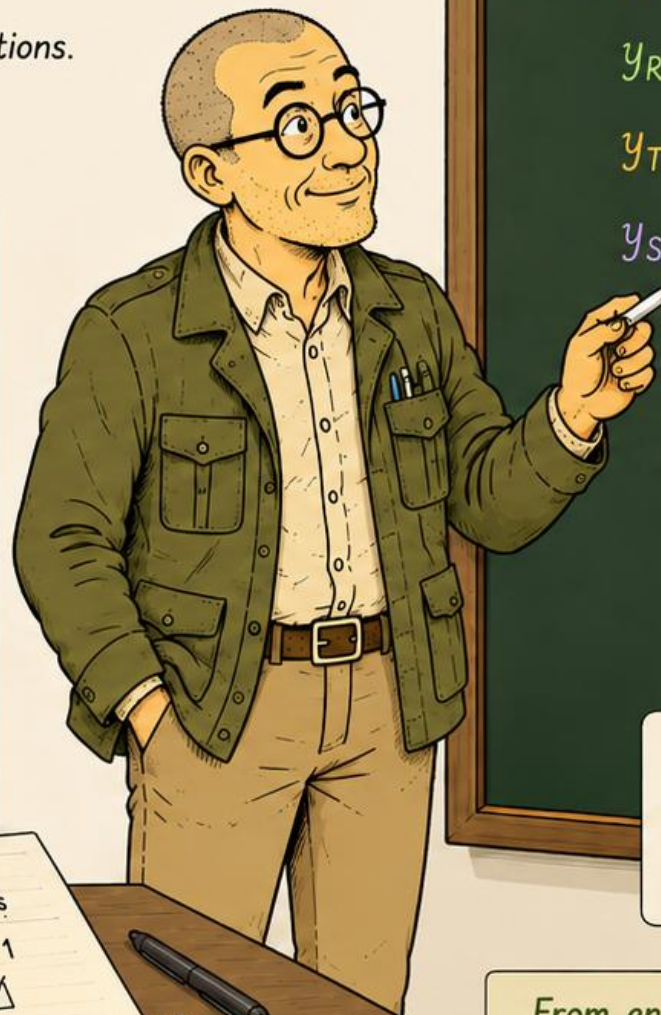
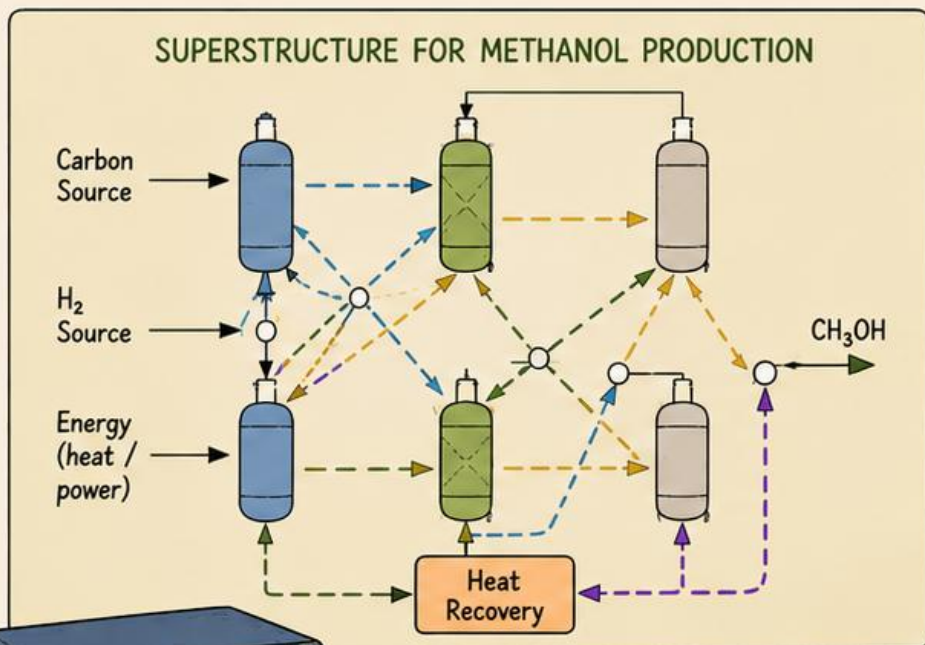


Core idea:

Preserve alternatives until optimization selects among them.

ENGINEERING CHOICES BECOME MATHEMATICS.

Our superstructure contains optional units and connections.
We represent each engineering choice with a *binary decision variable*.



WE DEFINE BINARY DECISION VARIABLES

y_H = Electrolysis

$y_H \in \{0,1\}$

y_R = Use of renewables

$y_R \in \{0,1\}$

y_T = Two-stage reactor

$y_T \in \{0,1\}$

y_S = Enhanced separation

$y_S \in \{0,1\}$

= Heat integration

$y_I \in \{0,1\}$

EXAMPLE OF ENGINEERING LOGIC

$$y_R \leq y_H$$

Renewables can only be used if electrolysis is selected.



The superstructure keeps all reasonable alternatives open.

The optimization will decide: 0 (off) or 1 (on).

From engineering possibilities to mathematical decisions.

PROCESS SYSTEMS
ENGINEERING

MATHEMATICAL
MODELING

OPTIMIZATION

PSE
LAB

Engineering
alternatives



Decision
variables

0 or 1



THE DESIGN SPACE.

EVERY DOT IS A POSSIBLE PLANT.

From all combinations of our decisions, we obtain a set of *feasible* configurations.

Each configuration has a *cost* and an *emissions* performance.

This is our *design space*.

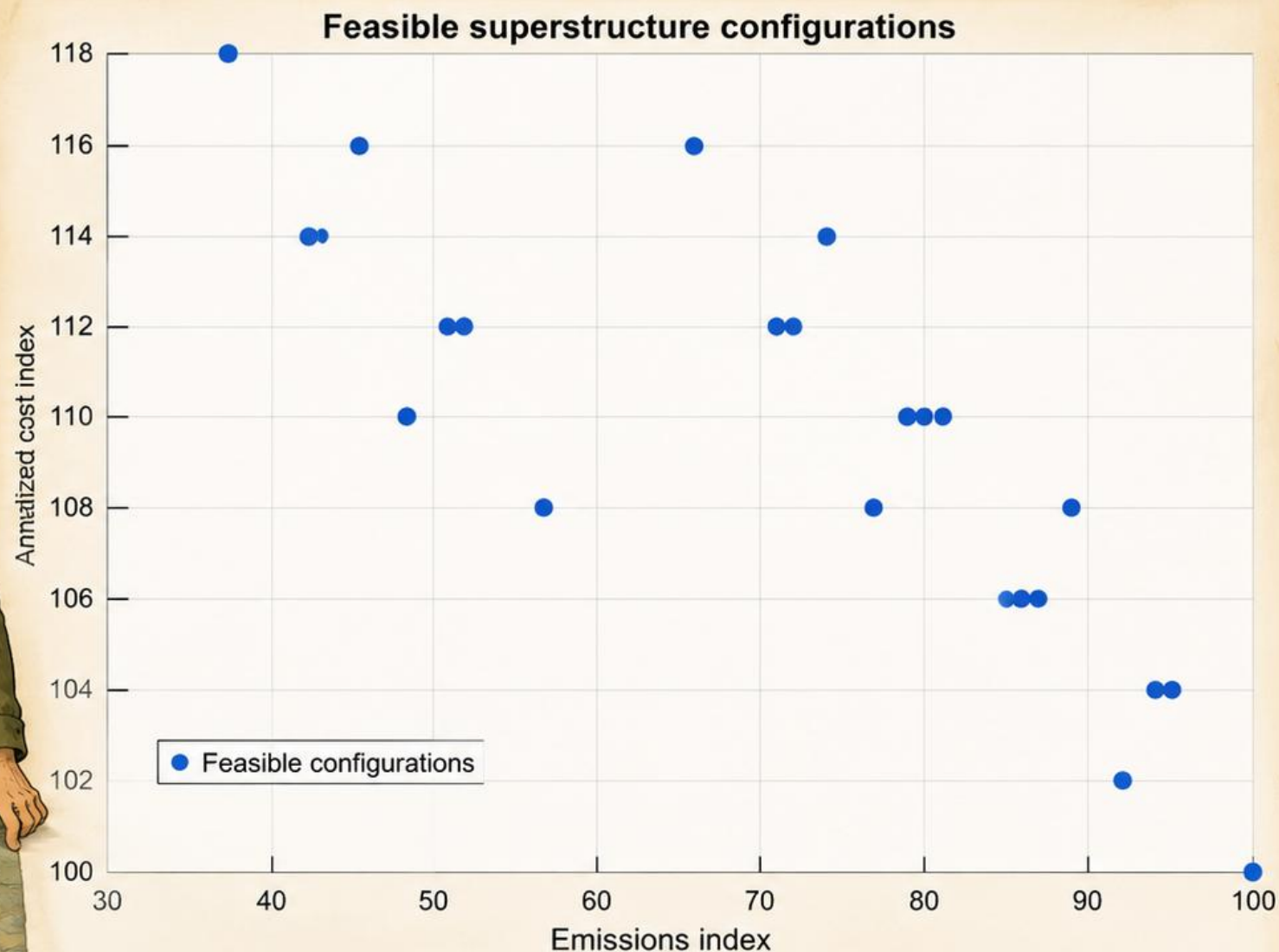


POSSIBLE PLANTS

- Cost
- Emissions
- Decisions
- Alternatives
- ...



We haven't chosen yet.
We're just exploring
what's possible.
The landscape is wide.
Which path is best?



AND THE WINNER IS...

Our optimization objective:
minimize cost.

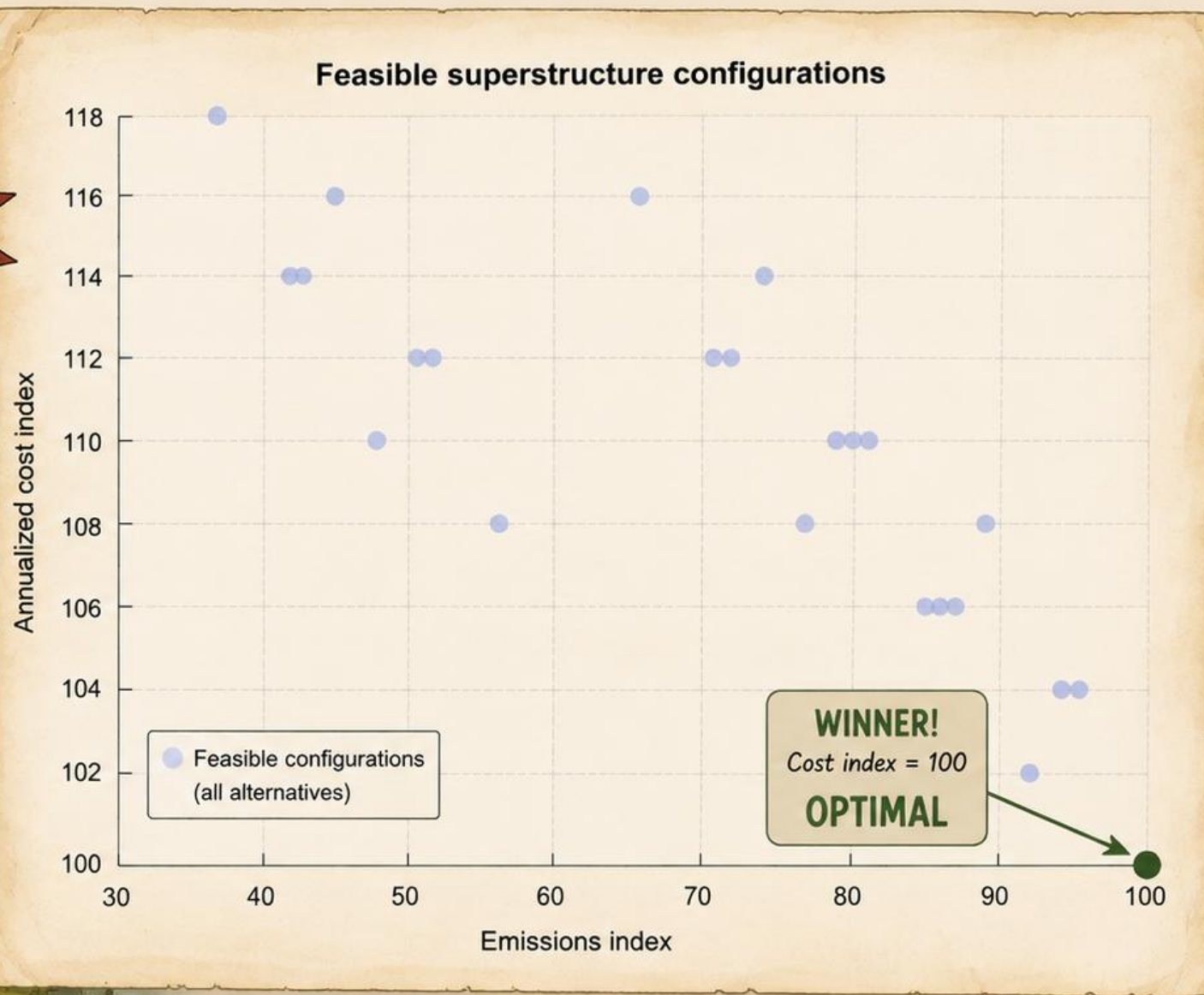
The optimization selects the
feasible configuration with
the *lowest cost.*

It uses *purchased H_2* and
a simple configuration –
no expensive low-carbon
options.



It is feasible.
It is cheaper than
every alternative.
Mathematically,
there is no ambiguity.

OPTIMAL.



We solved the problem... or did we?

OPTIMAL ACCORDING TO WHAT?

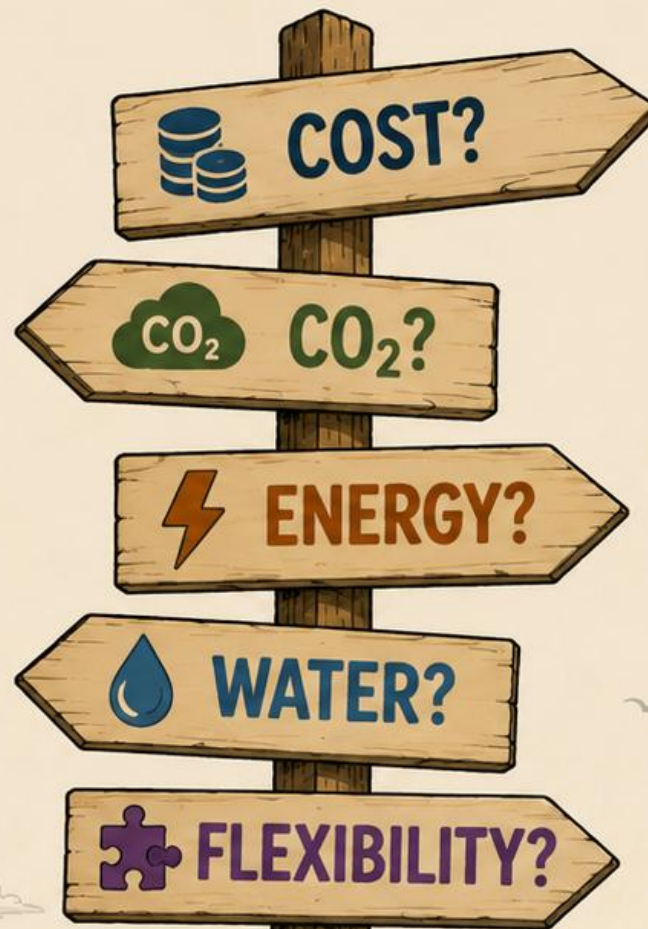
We optimized for *cost*.
But that was *our* choice,
not the optimizer's.



Optimization needs
an *objective*.
But which objective
should we choose?



The optimizer didn't decide
what mattered.
WE DID.



Different objectives
lead to different
"optimal" solutions.

Which ones matter
most for *us*?

OPTIMUM



CHANGE THE OBJECTIVE, CHANGE THE PLANT.

The optimizer gives us the best solution for the objective *we* choose.

Different objectives, *different winners*.



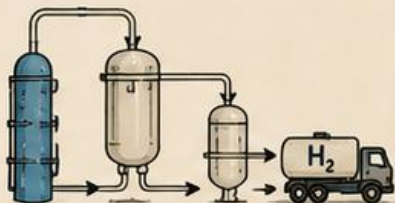
FOLLOW THE COST PATH

MINIMUM COST

Cost index = 100

Emissions index = 100

- Simple configuration
- Enriched H_2
- No expensive low-carbon options



WHICH ONE IS
OPTIMAL?

BOTH.



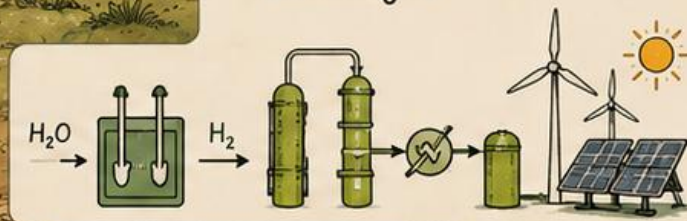
FOLLOW THE EMISSIONS PATH

MINIMUM EMISSIONS

Cost index = 118

Emissions index = 38

- Low-carbon configuration
- Electrolysis for H_2
- Renewables (wind / solar)
- Heat integration and other low-carbon technologies



We can optimize for different things.
The solution changes.

Cost optimum and emissions optimum are both mathematically optimal – with respect to *different objectives*.

BEYOND A SINGLE OPTIMUM.

When we optimize more than one objective, there isn't one best answer.

There is a **frontier** of trade-offs.

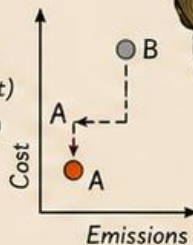
Multi-objective optimization

$$\min_{x,y} (C(x,y), E(x,y))$$

Minimize cost C and emissions E

DOMINANCE

Configuration A dominates B if A is **both** cheaper (lower cost) **AND** cleaner (lower emissions) than B.



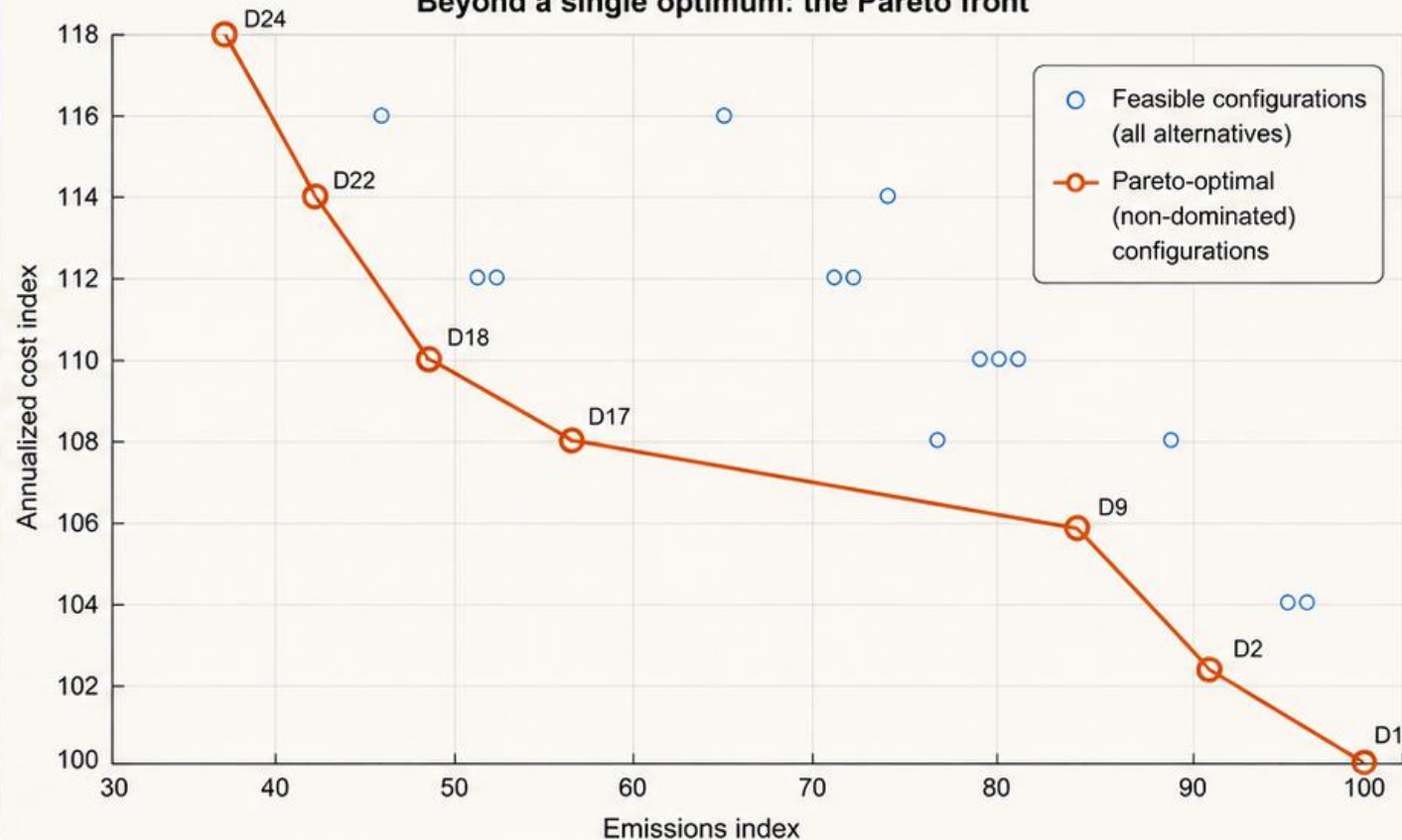
Dominated designs are unattractive.

These are the non-dominated (Pareto-optimal) configurations.



The **orange** designs are Pareto-optimal: improving one objective means worsening the other.

Beyond a single optimum: the Pareto front



Instead of producing a single answer, optimization reveals the **structure of the trade-off**.

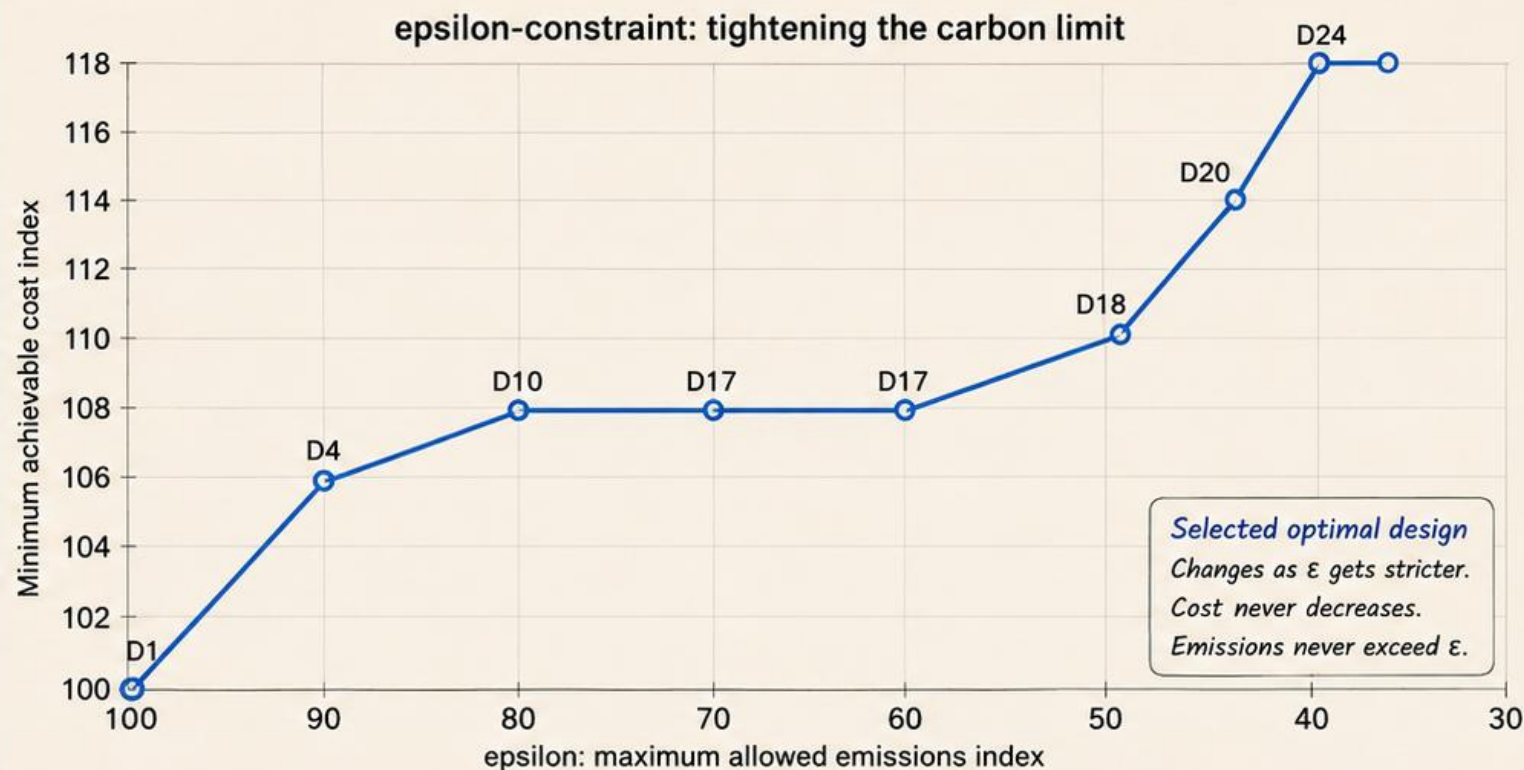
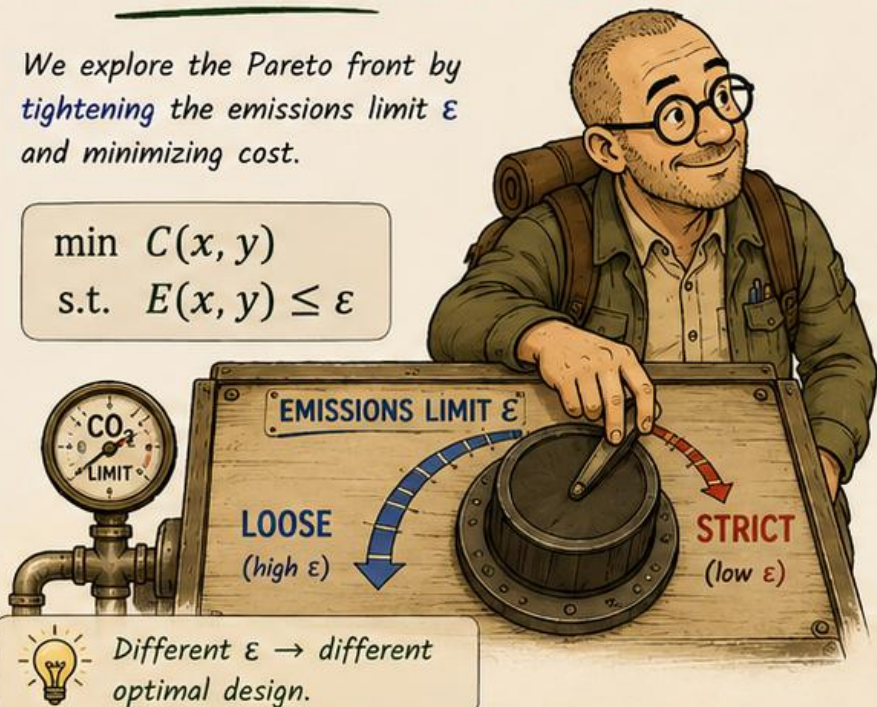


From the cheapest design (D1) to the cleanest design (D24), there are **many attractive compromises** in between.

TURN THE CARBON KNOB.

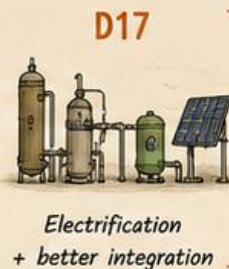
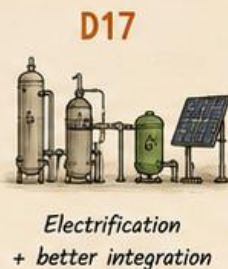
We explore the Pareto front by tightening the emissions limit ϵ and minimizing cost.

$$\begin{aligned} \min \quad & C(x, y) \\ \text{s.t.} \quad & E(x, y) \leq \epsilon \end{aligned}$$

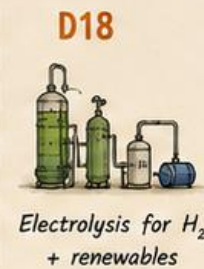


Optimal design
(on Pareto front)

Plant configuration
(technologies enter
as ϵ tightens)



Same
design
optimal
over a
range
of ϵ .



We do not just change a number. We **change the plant.**

OPTIMIZATION GIVES US CHOICES.

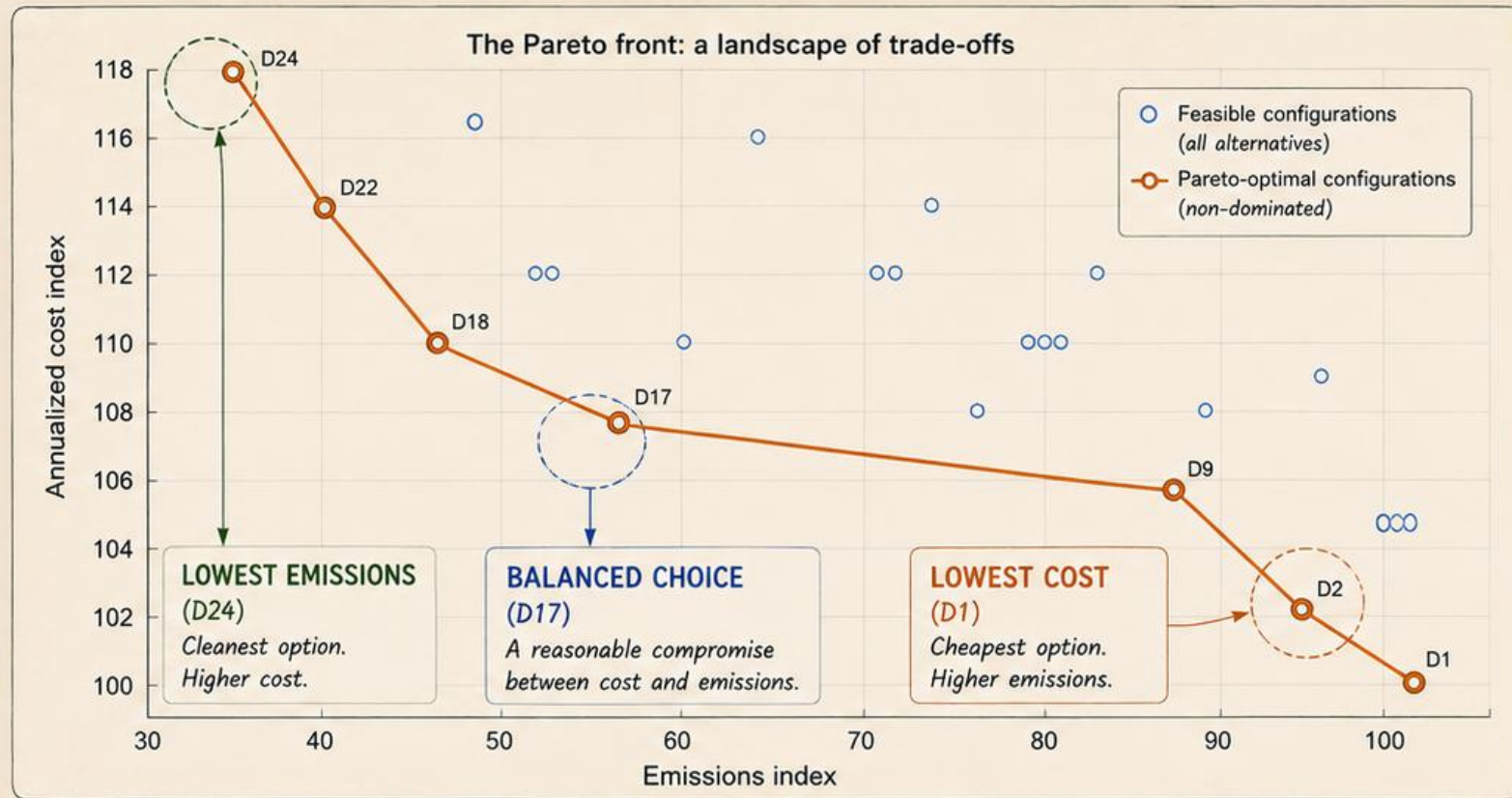
Multi-objective optimization does *not* give one answer. It reveals a *set of attractive alternatives* – each with consequences.

Different *objectives*.
Different *priorities*.

Different – but
rational –
decisions.



Use the *insights*,
choose *consciously*,
and adapt when
the world changes.



Low-emissions design (D24)



Cost index 118 Emissions index 38

Best for climate targets,
regulations, or green leadership.

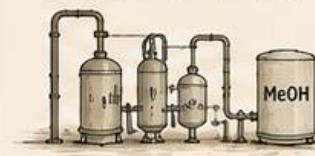
Balanced design (D17)



Cost index 108 Emissions index 60

Balanced performance –
a pragmatic compromise.

Low-cost design (D1)



Cost index 100 Emissions index 100

Best for short-term economics
or cost-driven decisions.

How is the final
choice made?

- Management priorities
- Carbon policy / limits
- Risk and uncertainty
- Future flexibility



Optimization has shown us the *choices* – and the *consequences* of those choices.

WE MADE ONE MORE ASSUMPTION...

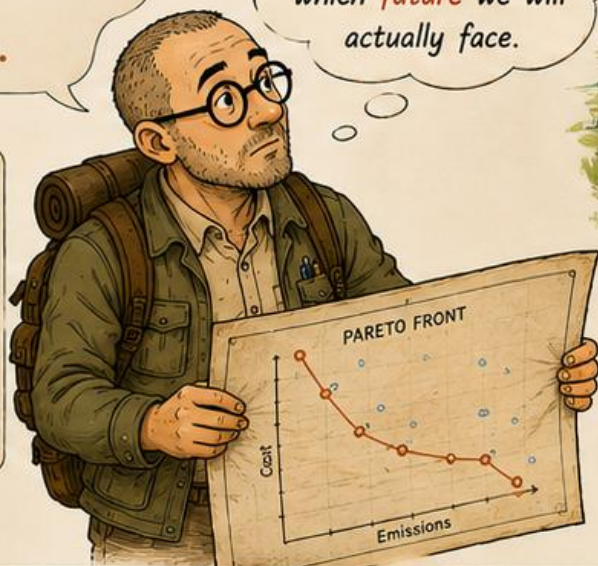
Up to now, we assumed we know the future.

But we don't.

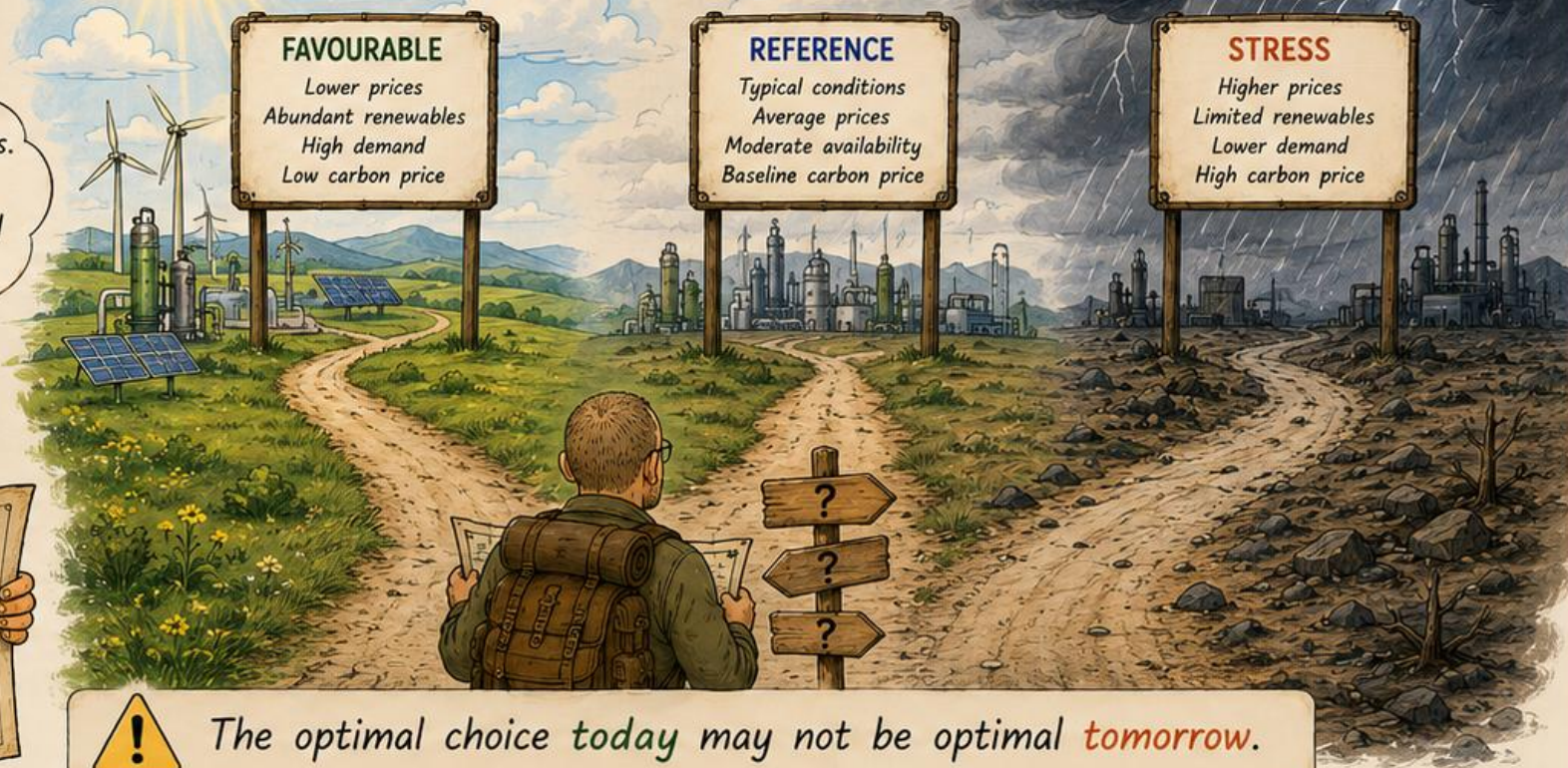
All our calculations used single values for prices, availability, demand, and model parameters.

In reality, the future can unfold in many ways.

We mapped the choices.
But we don't know which *future* we will actually face.



The future is *uncertain*.



The optimal choice today may not be optimal *tomorrow*.

Examples of *uncertain* factors we have not (and cannot) predict perfectly:



Purchased H₂ price

Markets, geopolitics, technology and policies drive large swings.



Electricity price

Demand, fuel prices, and markets cause volatility.



Renewable availability

Weather varies. Sun, wind and hydro are not guaranteed.

And many more...

- Demand
- Carbon price
- Feedstock price
- Process parameters
- Technology performance



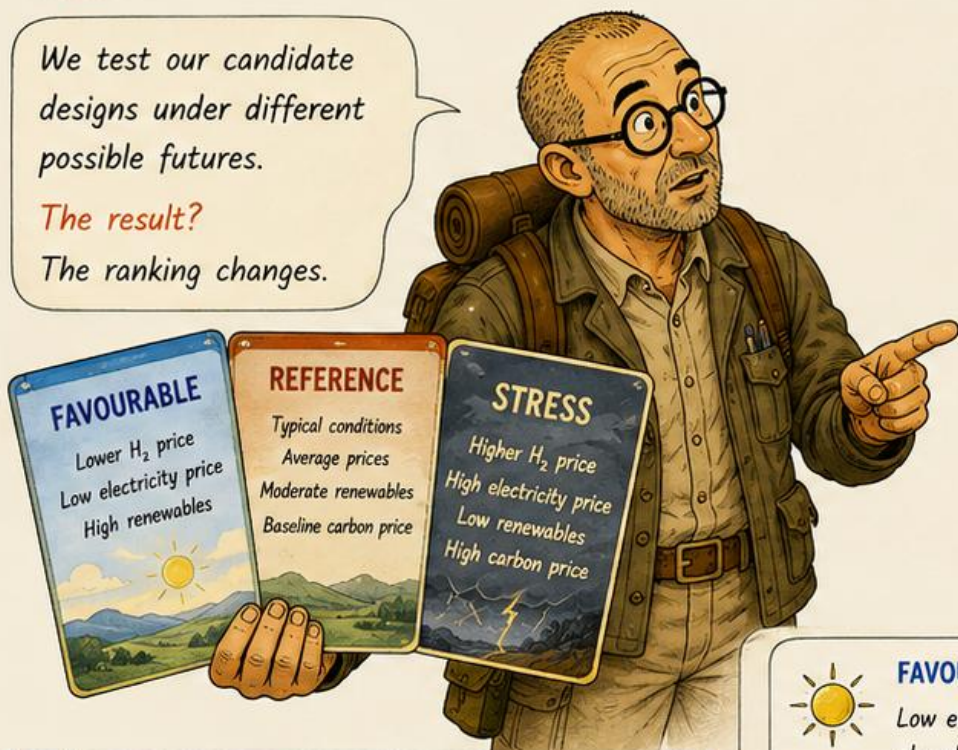
Optimization showed us the choices. Now we must face the *uncertainty*.

THE OPTIMUM MOVES WHEN THE FUTURE CHANGES.

We test our candidate designs under different possible futures.

The result?

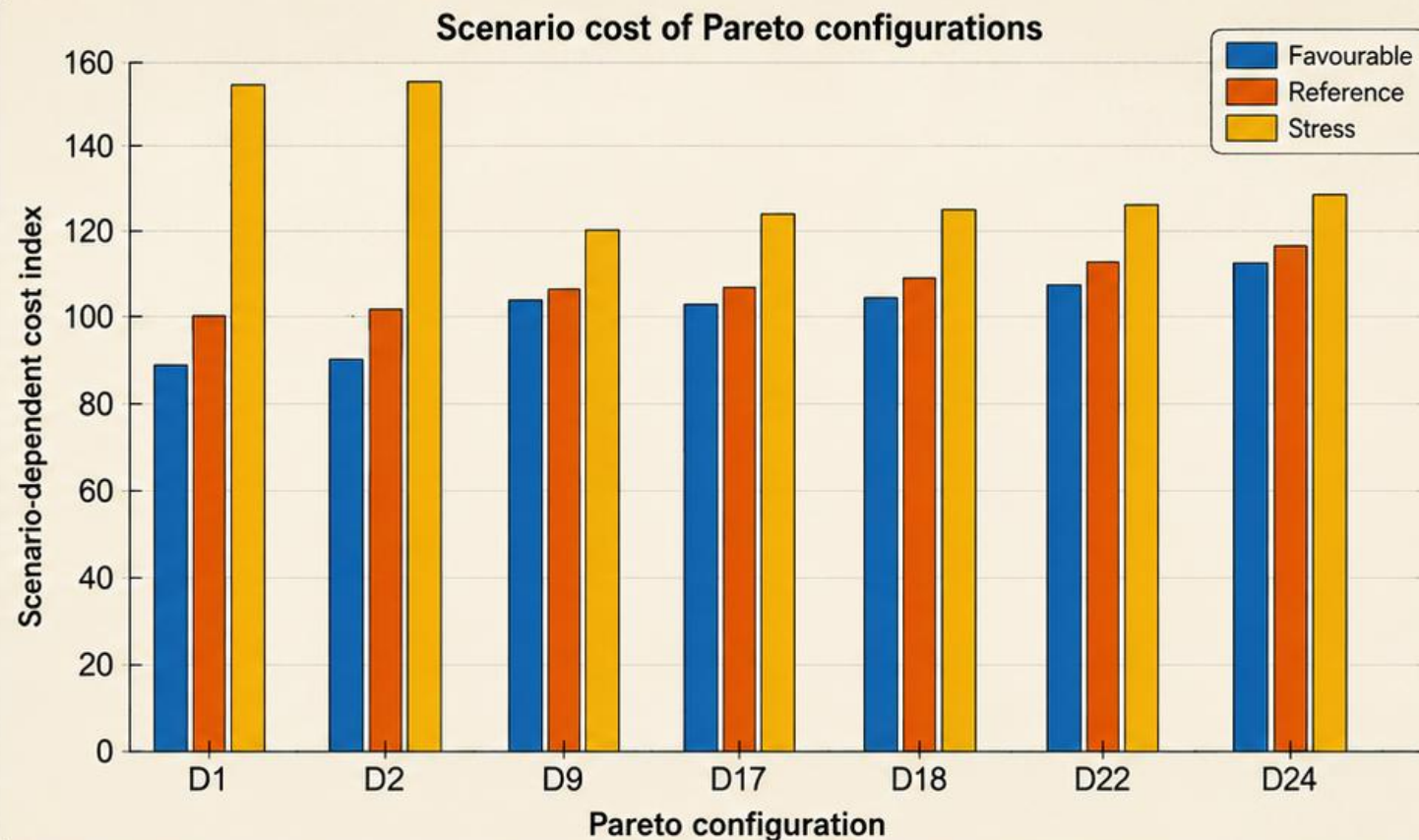
The ranking changes.



Different futures → different winners.

No single design is *best* in all futures.

We need solutions we can live with.



FAVOURABLE FUTURE

Low energy & H₂ prices, abundant renewables.

→ Lowest cost: D1



D1 and D2 look excellent.
D9–D24 are less competitive.



REFERENCE FUTURE

Typical conditions, our baseline view.

→ Lowest cost: D1



D1 is best (cheapest).
Others cost more.



STRESS FUTURE

High prices, scarce renewables, high carbon.

→ Lowest cost: D9



D1 and D2 become very expensive.
D9 (and others) perform much better.

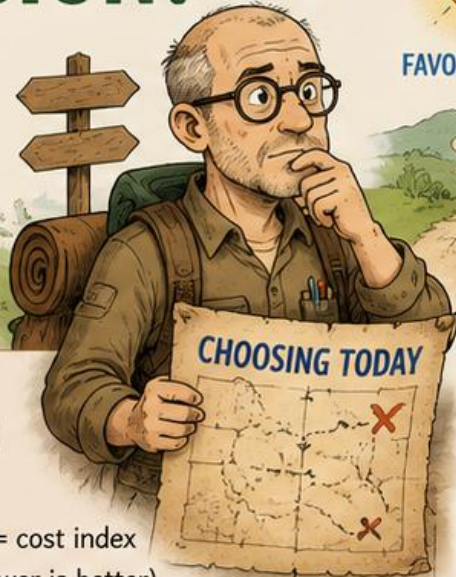


The design that looks optimal *today* may not be the design we wish we had built *tomorrow*.

WILL WE REGRET OUR DECISION?

We choose a design **today**, before knowing which future will occur.

If the future turns out differently, we may **wish** we had chosen another.



Regret in scenario for a chosen design d is the **extra cost** compared to the **best** design in **hindsight**.

$$R_{d,s} = C_{d,s} - \min_j C_{j,s}$$

C = cost index
(lower is better)

We look at the **worst regret** of each design across all scenarios ...

$$R_d^{\max} = \max_s R_{d,s}$$



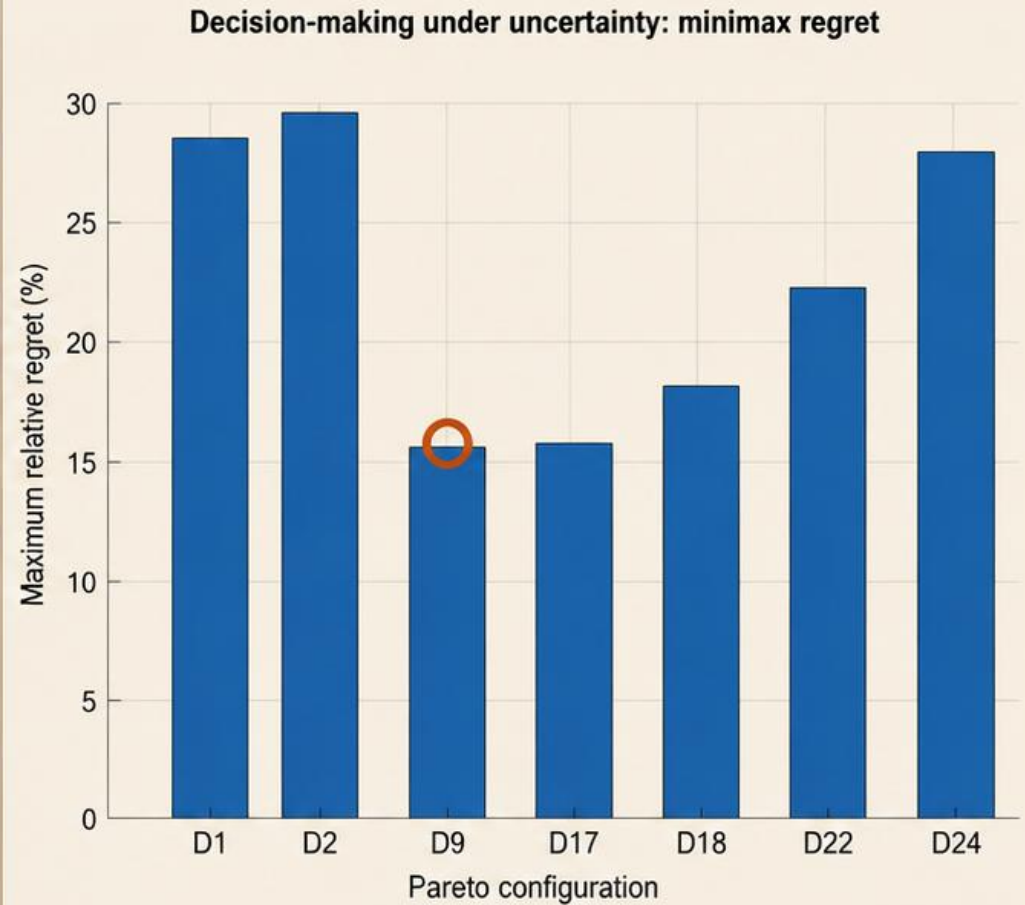
Minimax regret: choose the design with the smallest worst regret. A **robust compromise**.

How to read the figure

Lower bars = lower worst regret across all futures.

The circled bar is the minimax-regret solution in our example.

It **does not win** in every scenario, but it limits how much we could regret under the worst one.



Minimax regret does **not** guarantee the lowest cost. It aims to avoid the **biggest disappointment**.



Don't ask only:
"Which design wins?"



Ask instead:
"Which decision can we live with when the future surprises us?"



A good decision **today** is one we are unlikely to deeply regret **tomorrow**.

Optimization **informs** the decision.
Judgment makes the decision.



FROM MODELS TO DECISIONS

The complete chain: numbers → insight → judgment → impact

We connect the technical world to the real world of decisions.

1 MODELS

How does the system behave?



Mechanistic models

- Thermo, kinetics, transport
- First-principles understanding

Data-driven models / ML

- Learn from data
- Capture complex patterns

Hybrid modeling

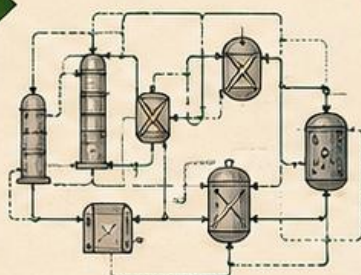
- Combine physics + data
- Better accuracy, wider range, faster evaluation



The workshop theme:
Hybrid modeling & ML
make our models richer.

2 SUPERSTRUCTURE

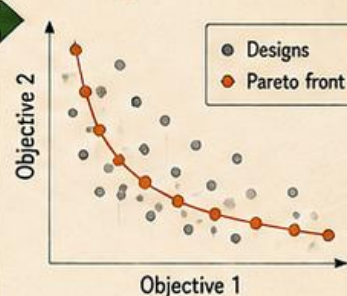
What alternatives are physically possible?



- All relevant units
- Multiple configurations
- Recycle, integration, process intensification
- The design space we can explore

3 OPTIMIZATION

Which configurations perform best?



- Mathematical programming (MILP / NLP)
- Pareto fronts
- Performance trade-offs
- Many good solutions, not a single best one

4 OBJECTIVES

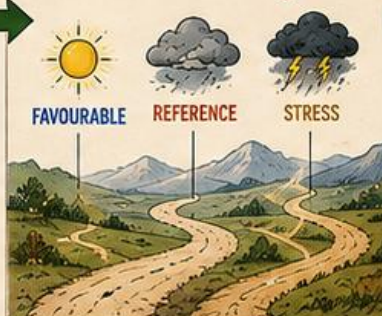
Best according to what?



- Cost / Affordability
- Emissions / Sustainability
- Energy / Efficiency
- Flexibility / Resilience
- Reliability / Safety
- ... and more (what we value matters!)

5 UNCERTAINTY

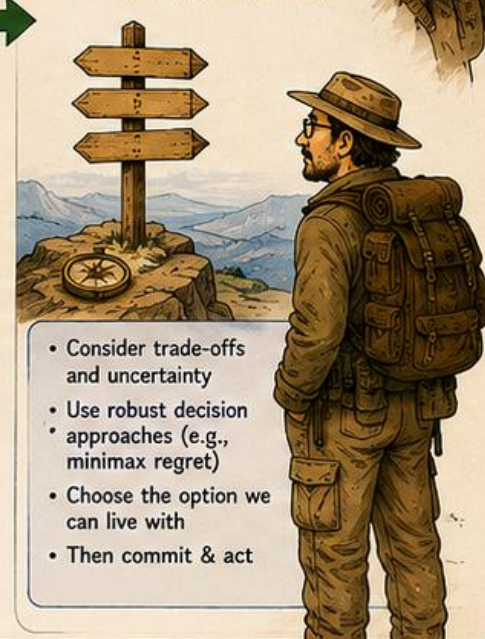
What happens when the future changes?



- Prices: H₂, electricity, CO₂
- Renewable availability
- Demand & operation
- Model & data uncertainty
- Scenarios / probabilities
- What we cannot know — but must prepare for

6 DECISION

Which alternative should we choose?



- Consider trade-offs and uncertainty
- Use robust decision approaches (e.g., minimax regret)
- Choose the option we can live with
- Then commit & act



Better models reduce error.



Bigger design space captures more options.



Optimization reveals performance & trade-offs.



Objectives reflect our values.



Uncertainty tests our resilience.



Decision creates real-world impact.



*The journey does not end with a better **model**.
It ends with a better **decision**.*

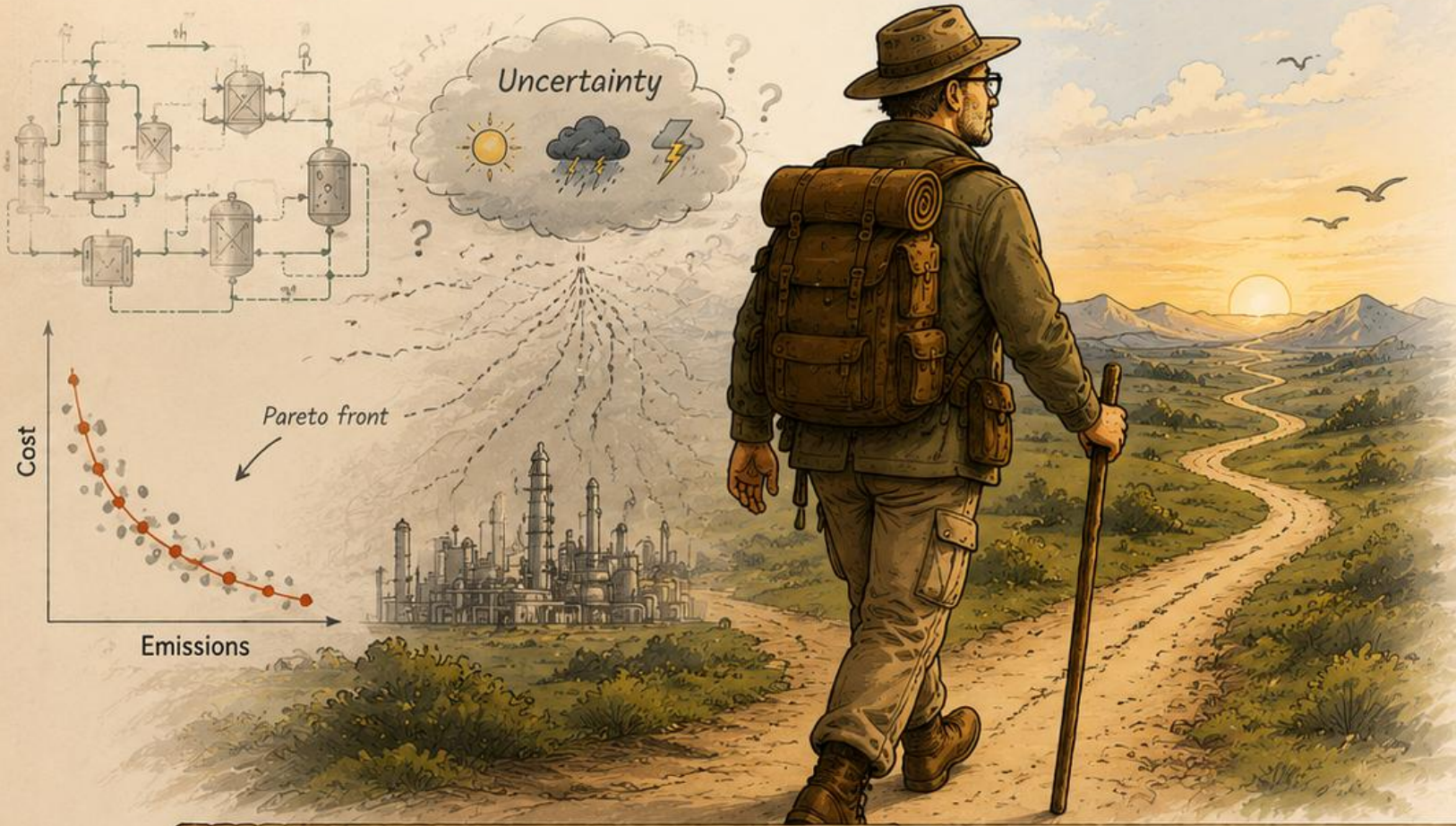
WORKSHOP THEME
IN ACTION



Hybrid modeling & ML improve every step in the chain. But their ultimate value is measured in **better decisions** under uncertainty.

BEYOND THE OPTIMUM

Chemical Engineering Decisions with Multiple Objectives and Uncertainty



The **optimum** is an answer to a mathematical question.

Engineering requires a **decision**.



Models predict.



Optimization explores.



Engineers decide.

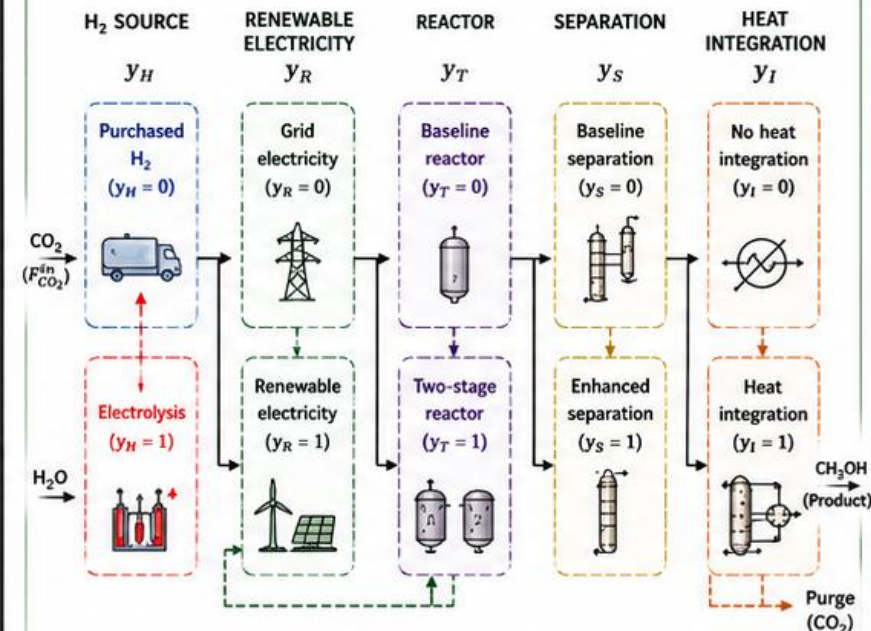


Better decisions **today** build a better **tomorrow**.



1. SUPERSTRUCTURE (FLOWSHEET CHOICES)

Binary decisions select technologies and options



Binary decision vector (technology selection)

$$\mathbf{y} = [y_H, y_R, y_T, y_S, y_I]^T, \quad y_i \in \{0, 1\}$$

Engineering logic (examples)

- Renewable electricity only if electrolysis is selected: $y_R \leq y_H$
- Two-stage reactor needs H_2 (always satisfied by both sources)
- Enhanced separation and heat integration are optional additions

2. PROCESS MODEL (PHYSICS + CONSTITUTIVE RELATIONS)

Key balances and constitutive equations (examples)

Reactor (stoichiometry: $CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$)

$$F_{CO_2}^{out} = F_{CO_2}^{in} - \xi$$

$$F_{H_2}^{out} = F_{H_2}^{in} - 3\xi$$

$$F_{MeOH}^{out} = F_{MeOH}^{in} + \xi$$

$$F_{H_2O}^{out} = F_{H_2O}^{in} + \xi$$

$$\xi = X_{CO_2}(y_T) F_{CO_2}^{in}$$

$$X_{CO_2}(y_T) = X_1 + (X_2 - X_1) y_T$$

(conversion increases with two-stage reactor)

Separation and recycle

$$F_i^r = \rho_i(y_S) F_i^{out} \quad (i = CO_2, H_2, MeOH, H_2O)$$

(recycle fraction depends on separation option y_S)

Hydrogen supply (topology enforced by binaries)

$$0 \leq F_{H_2, el} \leq M_{H_2} y_H \quad (\text{electrolysis if } y_H = 1)$$

$$0 \leq F_{H_2, pur} \leq M_{H_2} (1 - y_H) \quad (\text{purchased } H_2 \text{ if } y_H = 0)$$

Energy balance (steady state)

$$Q_{ext} + W_{el, in} = Q_{proc} + W_{el, proc}$$

Heat integration

$$Q_{ext} \geq Q_{proc} - \eta_{HI} y_I Q_{rec}$$

(heat recovery Q_{rec} available only if $y_I = 1$)

Continuous variables (examples)

Flow rates, temperatures, pressures, extents, duties, areas, powers, recycle flows, etc. $\mathbf{x} \in \mathbb{R}^n$

3. MATHEMATICAL PROGRAM (WHAT WE SOLVE)

Mixed-integer nonlinear program (MINLP)

$$\min_{\mathbf{x}, \mathbf{y}} [C(\mathbf{x}, \mathbf{y}), E(\mathbf{x}, \mathbf{y})] \quad (\text{multi-objective})$$

subject to

$$\begin{aligned} h(\mathbf{x}, \mathbf{y}) &= 0 && \text{mass \& energy balances} \\ g(\mathbf{x}, \mathbf{y}) &\leq 0 && \text{operating \& design constraints} \\ l(\mathbf{x}, \mathbf{y}) &= 0 && \text{constitutive relations} \\ m(\mathbf{x}, \mathbf{y}) &\leq 0 && \text{equipment \& capacity limits} \end{aligned}$$

Engineering logic (examples)

$$y_R \leq y_H \quad (\text{renewables only with electrolysis})$$

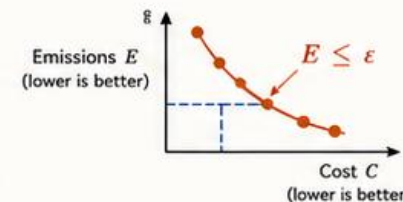
$$0 \leq y_i \leq 1, \quad y_i \in \{0, 1\} \quad (i = H, R, T, S, I)$$

ϵ -constraint form (to trace the Pareto front)

$$\min_{\mathbf{x}, \mathbf{y}} C(\mathbf{x}, \mathbf{y})$$

subject to all constraints above and

$$E(\mathbf{x}, \mathbf{y}) \leq \epsilon$$



Objectives (normalized indices)

$$C = \text{Total Annualized Cost Index} \quad [-]$$

$$E = \text{Emissions Index} \quad [-]$$



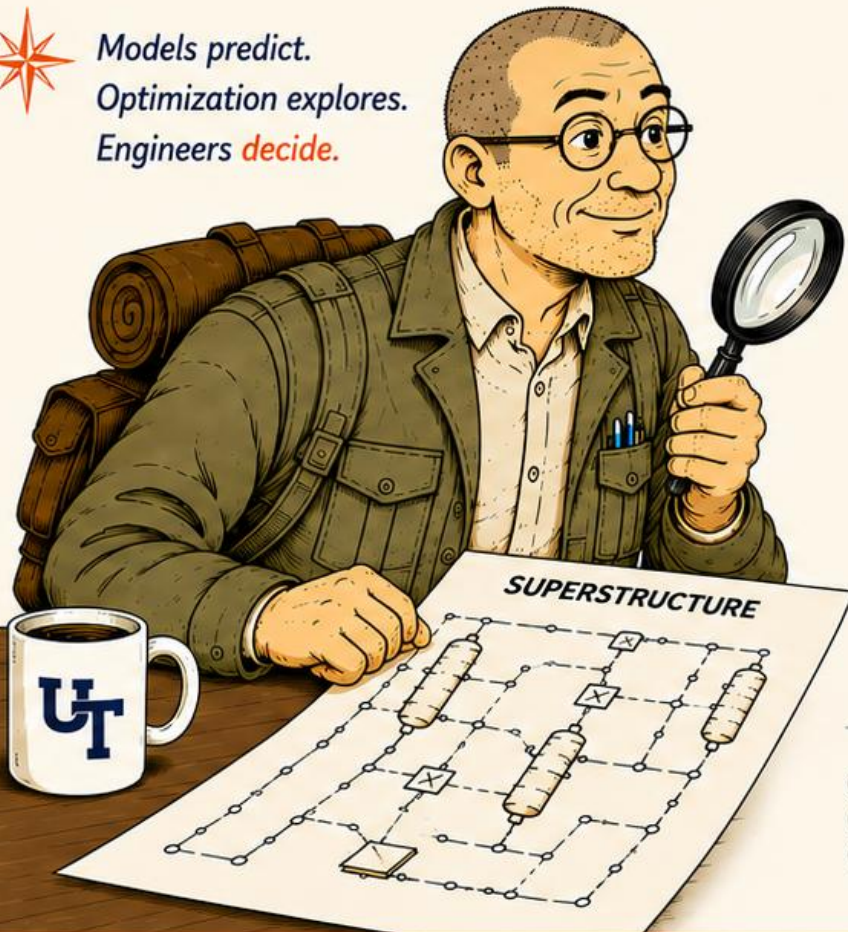
Binary decisions **choose** the flowsheet. Continuous variables determine how it **operates**.
Optimization does both **simultaneously**.

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*Chemical Engineering Decisions
with Multiple Objectives and Uncertainty*



Models predict.
Optimization explores.
Engineers *decide*.



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cost and emissions



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